

KAMAL PATEL

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EDUCATION

Rutgers, The State University of New Jersey · New Jersey September 2022 - June 2026
MS & Doctoral studies in Industrial & Systems Engineering (Completed two years of PhD courses) New Jersey
Courses: Production Analysis, Supply Chain Engineering, Linear Programming, **Lean Six Sigma (Certification)**
Regression Analysis, Systems Reliability, Data Mining II, Time-Series Analysis.

WORK EXPERIENCE

Graduate Teaching Assistant Jan 2025 – June 2026
Rutgers University - Industrial & Systems Engineering Piscataway, NJ

- Mentored 30+ students in applying linear programming and Excel Solver to solve complex supply chain network design and resource allocation problems.
- Facilitated weekly reliability analysis recitations and mentored students through MATLAB-based assignments.

Industrial Engineer Intern May 2023 - Aug 2023
Kismet Technologies Orlando, FL

- Customized Katana MRP software for multi-SKU inventory management, improving stock accuracy by 20%.
- Overhauled batch manufacturing data tracking by building custom PowerApps solution, driving an 11% boost in operational efficiency.
- Standardized production scheduling and batch acceptance testing workflows, achieving ISO 9001 compliance and reducing quality-related production delays by 15%.

Project Head Nov 2022 - Dec 2022
Mareana: Industry Consulting New Brunswick, NJ

- Spearheaded a 4-person student consulting team to analyze and identify bottlenecks in turbine manufacturing.
- Engineered a FlexSim discrete event model to optimize line balancing, reducing production cycle time by **23.5%**.

PROJECTS

E-Commerce Delivery Optimization Analytics | MySQL [GitHub](#)

- Analyzed 100K+ order fulfillment records to evaluate On-Time In-Full (OTIF) performance and pinpoint root causes of network-wide delivery latency.
- Developed CTE and window-function queries to rank logistics performance and isolate high-risk product categories by geographical region.
- Calculated \$213K in revenue at risk from high-priority order exposure using SQL CTEs, identifying a \$68.6K regional bottleneck that accounted for 32% of total value.

Supply Chain KPI & Fulfillment Analytics Dashboard | Power BI, DAX [GitHub](#)

- Consolidated 5 years of multi-category inventory data (640K+ units) in Power Query into one standardized model, enabling consistent tracking of inventory, fulfillment and logistics KPIs.
- Quantified inventory efficiency at 23.5 turns per year (15.6 days of sales on hand), showing stock cycles through in about two weeks and confirming a lean working-capital position.
- Discovered a 30.2% unfulfilled order rate, quantifying \$21.7M in order value at risk, identifying the West region as the primary area for inventory optimization.

Electricity Price Time Series Forecasting [GitHub](#)

- Engineered PyTorch LSTM and autoregressive models to forecast Locational Marginal Pricing (LMP), optimizing prediction accuracy through advanced time-series feature engineering.
- Validated model performance against traditional baselines, achieving an MAE of 3.92 and RMSE of 9.21 to outperform the benchmark SARIMAX model.

TECHNICAL SKILLS & LEADERSHIP

Languages: Python (Pandas, NumPy SciPy), R, SQL, MATLAB, DAX
Developer Tools: MySQL, Excel Solver, Power BI, FlexSim, GurobiPy, AnyLogic

ACHIEVEMENTS/CERTIFICATIONS

- Awarded **Best Project Performance** (Regression Analysis) and **Best Presentation** (Data Mining II) at Rutgers University.